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Threat Intelligence

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Introduction

In late 2025, Mandiant responded to a security incident involving a compromised web server running [KnowledgeDeliver](#). KnowledgeDeliver is a Learning Management System (LMS) developed by Digital Knowledge commonly used in Japan. Mandiant [identified a critical vulnerability](#) that allowed unauthenticated Remote Code Execution (RCE). An unknown threat actor leveraged this access to inject malicious code into the LMS platform, with the goal of infecting users visiting the site.

This vulnerability stems from the use of identical pre-shared ASP.NET machine keys across multiple customer deployments. The vulnerability was initially exploited as a zero-day, now tracked as [CVE-2026-5426](#).

The Vulnerability

KnowledgeDeliver installations deployed before Feb. 24, 2026 relied on a standardized `web.config` file provided by the vendor. This configuration file contained hardcoded `machineKey` values used by the ASP.NET framework to encrypt and sign data, including ViewState payloads.

Because these keys were identical across independent customer environments, a threat actor who obtained the keys from one deployment could compromise any other internet-facing KnowledgeDeliver instance.

The following is an example of the relevant configuration line found in the `web.config` file:

```
<machineKey decryptionKey="<REDACTED>" validationKey="<REDACTED>" />
```

The ASP.NET ViewState persists page state across postbacks. When the `machineKey` is known, a threat actor can craft a malicious ViewState payload. By sending this payload in an HTTP request (via the `__VIEWSTATE` parameter), the threat actor can make the server deserialize it.

This technique follows the pattern of the [ViewState Deserialization Zero-Day Vulnerability](#) affecting Sitecore (previously reported by Mandiant), and [Code injection attacks using publicly disclosed ASP.NET machine keys](#) reported by Microsoft. This highlights how it is critical to keep the machine key unique and secure.

Post-Exploitation Activity

Once access was established, the threat actors focused on maintaining their presence and expanding the impact of the compromise.

BLUEBEAM Web Shell Deployment

The threat actor deployed a .NET-based in-memory web shell called BLUEBEAM (also known as Godzilla). The use of BLUEBEAM is consistent with the [Microsoft reporting](#). This malware operates entirely in memory within the IIS worker process (`w3wp.exe`), making it

difficult to detect through traditional file-based scanning. It allows threat actors to execute further commands and payloads by sending encrypted data via HTTP POST request bodies.

File Tampering

The threat actor was observed executing commands to escalate their control over the web server's file system:

-

Permission Modification: The threat actor used `icacls` to grant "Everyone" full access to the web application directory.

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JavaScript Tampering: The threat actor modified an application JavaScript file, adding code to perform the following:

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Display a fake security alert, prompting users to install a "security authentication plugin".

-

Silently load a remote malicious script hosted on a threat actor-controlled domain.

Cobalt Strike Infection

The remote script convinced users to download a fake installer, which led to workstations being infected with a Cobalt Strike BEACON backdoor. The payload was encrypted using a key that used the name of the compromised organization, which indicated that the threat actor prepared this payload specifically for the targeted organization.

How to Hunt for This Activity

Organizations should monitor for the following indicators to identify potential ViewState exploitation and post-exploitation activity.

1. Application Event Logs (Event ID 1316)

Monitor the Windows Application log for Event ID 1316 from the source `ASP.NET 4.0.30319.0` (or similar).

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Failed Attempt (Integrity Failure): `Event code: 4009-+-Viewstate verification failed. Reason: The viewstate supplied failed integrity check.` May indicate an attack attempt with an incorrect key.

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Successful Execution (Invalid ViewState): `Event code: 4009-+-Viewstate verification failed. Reason: Viewstate was invalid.` Confirms integrity checks were passed. Deserialization of the payload was attempted and may have succeeded. The payload may or may not have been executed.

Mandiant decrypted payload strings recorded in the event log messages with the server's machine keys and recovered a payload related to a BLUEBEAM web shell.

2. Suspicious Process Activity

Monitor for unusual child processes spawned by `w3wp.exe`. Commands observed include:

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```
cmd.exe /c ...
```

-

```
whoami
```

-

```
powershell.exe
```

3. File Integrity Monitoring

Monitor for unauthorized changes to `.js`, `.aspx`, or `.config` files within the web root. Specifically, look for the addition of remote script loaders or unusual logic in commonly used libraries.

4. Anomalous User-Agent Strings

Mandiant identified User-Agent strings consisting of two distinct identifiers concatenated together, which were consistent with ones reported in [ViewState Deserialization Zero-Day vulnerability](#). Monitor for web request logs for such anomalous User-Agent strings. The following are examples of identified User-Agent strings:

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```
Mozilla/5.0 (Windows NT 6.1) AppleWebKit/537.2 (KHTML, like Gecko) Chrome/22.0.1216.0  
Safari/537.2 Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like  
Gecko) Chrome/121.0.0.0 Safari/537.36
```

-

```
Mozilla/5.0 (Windows; U; Windows NT 6.1; en-US; rv:1.9.2.13) Gecko/20101213 Opera/9.80  
(Windows NT 6.1; U; zh-tw) Presto/2.7.62 Version/11.01 Mozilla/5.0 (Windows NT 10.0;  
Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/121.0.0.0 Safari/537.36
```

-

```
Mozilla/5.0 (compatible; MSIE 9.0; Windows NT 6.1; Trident/5.0)  
chrome/10.0.648.205 Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36  
(KHTML, like Gecko) Chrome/121.0.0.0 Safari/537.36
```

Remediation and Mitigation

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Rotate Machine Keys: Immediately generate a unique, cryptographically strong machine key for each KnowledgeDeliver instance. This is the only way to invalidate the shared

secret.

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Restrict Access: If possible, limit access to the LMS to known organizational IP address ranges.

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Investigation: Hunt for this activity, and conduct a thorough investigation if any signs of exploitation are identified.

Outlook and Implications

The exploitation of KnowledgeDeliver highlights the severe risks of using shared secrets in deployment templates. A single leaked key can compromise an entire ecosystem of installations. By implementing unique secrets and robust endpoint monitoring, organizations can defend against these deserialization attacks.

Indicators of Compromise (IOCs)

To assist the wider community in hunting and identifying activity outlined in this blog post, we have included indicators of compromise (IOCs) in a [free GTI Collection for registered users](#).

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File Name

|

Type

|

SHA-256

|||

LoadLibrary.dll

|

BLUEBEAM

|

7c1f99dca8e5a7897892f9d224a6495023a2cfd2671697d229d355978c415ed2

||

Google Security Operations (SecOps)

The following [SecOps](#) searches can be used to hunt for this activity.

```
(metadata.log_type = "WINEVTLOG" or metadata.log_type = "WINEVTLOG_XML")  
metadata.product_event_type = "1316"  
additional.fields["Message"] = /Event code: 4009\b/ nocase
```

```
(metadata.event_type = "PROCESS_LAUNCH" or metadata.event_type = "PROCESS_OPEN") AND  
principal.process.command_line = /w3wp.exe/ nocase AND  
target.process.command_line = /cmd.+ \c |whoami|powershell/ nocase
```

SecOps customers have access to the following rules and more under the Mandiant Hunting Rules, Mandiant Frontline Threats, Mandiant Intel Emerging Threats rule packs:

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ASP.NET ViewState Deserialization Attempt

-

W3wp Launching Cmd With Recon Commands

-

W3wp Launching Encoded Powershell

-

W3wp Launching Icacs

-

Web Server Process Launching Whoami

-

IIS ViewState Exploitation Success

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IIS ViewState Exploitation Followed by Web Root File Tampering

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Possible Windows Exchange Server Spawning Shell

Acknowledgements

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